

V-Beta Overview

Product Overview

V-Beta is a web platform designed to capture, organize, and share climbing beta within indoor bouldering communities. In climbing, beta refers to the movement sequences, techniques, and strategies used to complete a specific climbing problem.

Indoor climbing gyms frequently introduce and remove climbing problems as part of routine wall resets. Knowledge about how to complete these problems is often shared informally through conversation or temporary media, making it difficult to preserve, discover, or compare solution techniques over time.

V-Beta provides a structured system where climbers can record and share solution demonstrations, discuss strategies, and contribute feedback on perceived difficulty. The platform enables multiple solution approaches to be documented for each climbing problem, creating a searchable and collaborative knowledge repository.

The platform also supports route setters and gym organizations by providing tools to manage climbing problems, track wall resets, and notify climbers when new problems become available. By connecting climbers, route setters, and gyms within a unified system, V-Beta improves knowledge retention, encourages community collaboration, and enhances engagement with climbing facilities.

Core Domain Concepts

Gym

A **Gym** represents a physical climbing facility.

Responsibilities:

- Owns wall sections and climbing problems
- Defines naming conventions for wall sections
- Hosts a community of climbers and route setters

Invariants:

- A climbing problem belongs to exactly one gym
- Wall sections are unique within a gym

Wall Section

A **Wall Section** represents a fixed physical area of a gym wall.

Attributes:

- Name (gym-defined)
- Wall angle (Slab, Vertical, Overhang, Roof):
 - **Vertical:** Relatively straight up and down, climbers utilize strength and technique abilities.
 - **Overhang:** Leans at an angle steeper than vertical, adds an increased need for body tension and power.
 - **Roof:** Steepest wall angle, often nearing horizontal – calls for controlled core strength and increased pulling footwork.
 - **Slab:** Gentle angle that leans slightly away, taps into balance and precise footwork.
- Optional angle metadata (e.g., degrees)

Invariants:

- Wall sections cannot be deleted while they contain active problems

Climbing Problem (Aggregate Root)

A **Climbing Problem** is a climbing challenge installed on a wall section.

Owned by: Gym

Located in: One Wall Section

Attributes:

- Assigned difficulty grade (V-scale)
- Hold description (e.g., color or set)
- Location within wall section
- Creation date

Lifecycle States:

- Draft – defined but not visible to climbers
- Active – available for climbing and beta
- Inactive – removed due to wall reset or retirement

Invariants:

- Only route setters may create, activate, or deactivate problems
- An inactive problem cannot receive new beta
- A problem must have exactly one difficulty grade while active

Wall Reset (Domain Event)

A **Wall Reset** represents the removal of all problems from one or more wall sections.

Effects:

- All affected problems transition to Inactive
- Beta remains readable but locked

User Roles

Climber

A **Climber** is any user who can:

- View climbing problems
- Submit beta content
- Suggest alternative difficulty ratings
- Participate in discussions

Route Setter

A **Route Setter** is a climber with additional authority:

- Create and manage climbing problems
- Assign official difficulty grades
- Perform wall resets

Beta Content

Beta Content represents a climber's proposed solution to a climbing problem.

Attributes:

- Media (video, optional text)
- Author (climber)
- Creation timestamp

Rules:

- Multiple beta entries may exist per problem
- Beta is immutable after creation
- Beta belongs to exactly one climbing problem

Difficulty Grade (Value Object)

A **Difficulty Grade** is a standardized value in the V-scale: VB, V0 ... V17

Rules:

- Only route setters assign official grades
- Climbers may submit grade suggestions, stored separately